

Reducing Measurement Error through Questionnaire Design (and Pretesting)

Surv/SurvMeth 621

Fundamentals of Data Collection

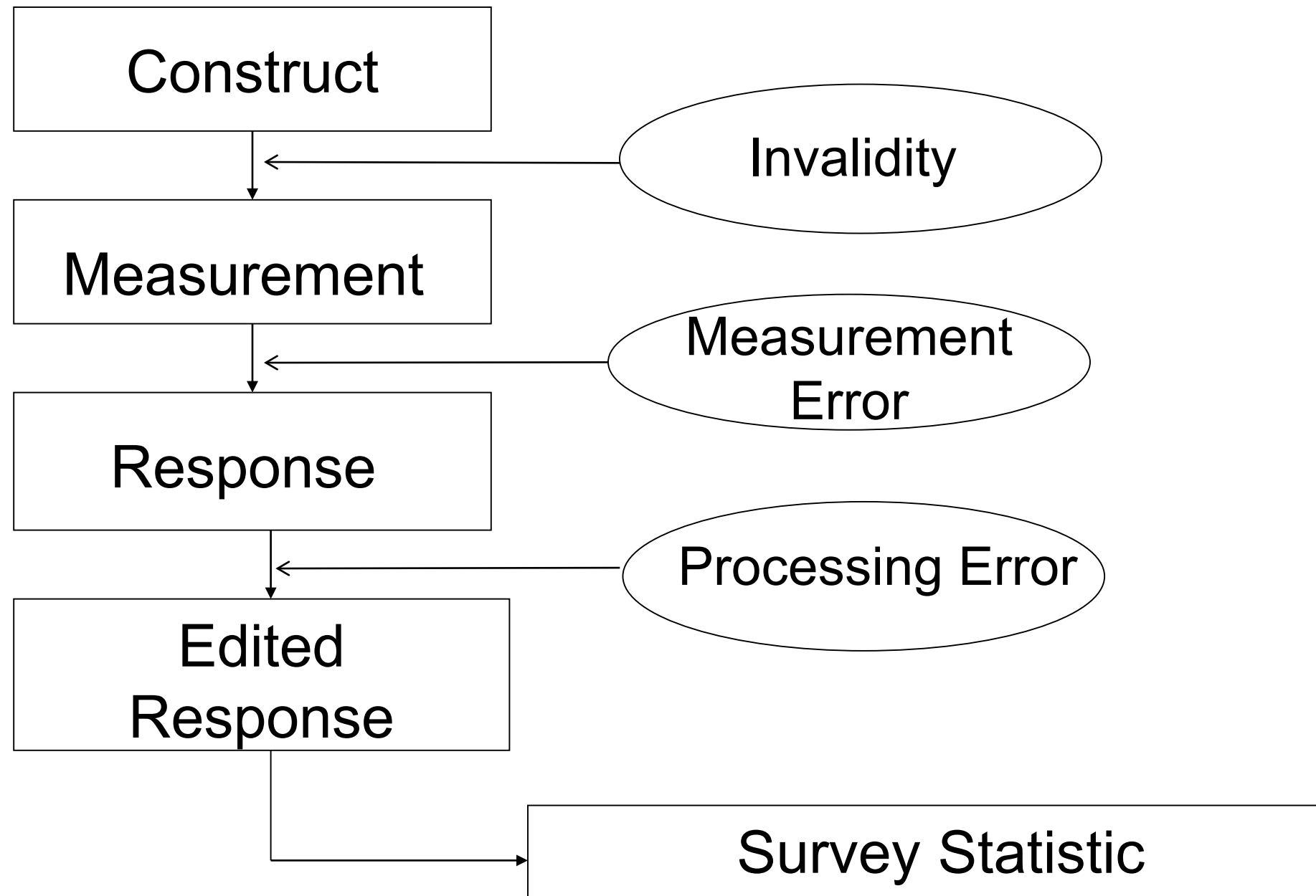
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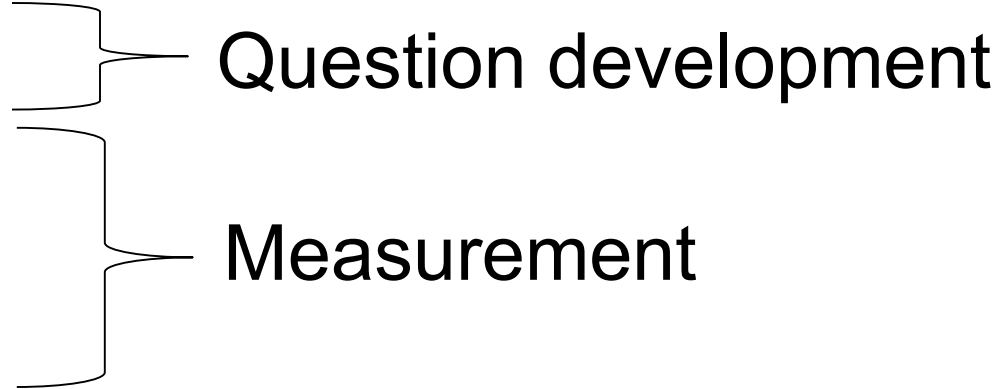
Outline

- Goals of questionnaire design
- Questions about facts and behaviors
 - Guidelines for design of questions of various types
- Questions about subjective phenomena
 - Guidelines for design of questions of various types
- Pretesting

The survey life cycle



Goals of Questionnaire Design Process

- To develop measures that are:
 - Valid (successfully operationalize construct)*
 - Reliable (produce consistent measurement)
 - Standard (comparable across people)
 - Easy to administer (practical)
- 
- Question development
- Measurement

*Not to be confused with use of “validity” to mean accuracy

Validity

- Define research/measurement objectives
 - What is the (analytic) purpose of each question?
 - What information needs to be measured?
 - How will the data be analyzed? How will the results be reported?
- Develop question wording that will fulfill objectives
 - Does the question capture the construct you wish to measure?
 - Relies on intuition, pretesting, existing questions
 - Not about measurement error per se – but extent to which each question measures corresponding concept
 - May take > 1 Q to fulfill a particular objective

Anticipating How the Data Will be Used

To what extent do you agree or disagree with the following statements about the impact of technology on your work?

A. Technology has removed some of my tasks

B. Technology has created new tasks

C. Technology has changed some tasks in my work

- What are the likely objectives of this question?
- What kind of data will it produce? How would you analyze them?
- How would you report/display the results?

Validity example

- Objective: Measure Job Satisfaction
 - If we assume job satisfaction depends at least on duties, compensation, environment, not enough to ask, “How satisfied are you with your job?”
 - Q1a: How satisfied are you with the work you do as part of your job?
 - Q1b: How satisfied are you with the amount of money you are paid?
 - Q1c: How satisfied are you with the facilities in which you work?
- Designer might question whether “satisfaction” is exactly the right dimension and iterate with researcher

Types of Validity

- Content or face validity
 - On the face of it, Q seems to be measuring what its intent to measure
- Criterion validity
 - Correlation with another variable with which should be associated, often in same data set
 1. *Concurrent validity*: both variables measured at same time, e.g., party affiliation and ideology collected in same questionnaire
 2. *Predictive validity*: variable measured in one wave of a survey, e.g., preference for a candidate, predicts a variable measured in a later survey, e.g., reported voting behavior
- Construct validity
 - Correlation between intended measure and proxy measure, e.g., amount of education as proxy for cognitive ability
- Comparison with gold standard such as administrative records, e.g., voter registration lists, or sensor measures, e.g., step count

Measurement Error: Reliability

- Answer consists of true “score” + error:

$$Y_{it} = T_i + \xi_{it}$$

- Reliability (Classical test theory version): What proportion of observed variance is variation in the true score?

$$\rho = \sigma_{true}^2 / \sigma_{obs}^2$$

- If true scores vary with observed scores, measurement is *relatively* accurate, i.e., actual scores may differ but “move together”
 - but not necessarily accurate in an absolute sense
- In practice, “reliability” means agreement between observable measures, e.g., same sample at 2 times or 2 groups at same time

Reliability (cont' d)

- How to measure reliability?
 - Test-retest
 - Split half
 - Related measures (Cronbach' s alpha)
 - Simple response variance (inversely related to reliability)
- In practice, reliability is cheaper and easier to measure than validity
 - e.g., Simple response variance is always present in data but administrative records are often not available
- But, it is less informative about data quality than validity
 - Relative versus absolute accuracy

Standardization

- Is everyone answering the same question?
- Closely related to reliability
 - If interviewers trained to adhere to the script, variation across interviewers should be minimal
- Scripting the interaction via the questionnaire increases standardization and thus comparability of the answers, relative to unstructured questionnaire
- One consideration in designing standardized measurement: is the number of response options manageable by all *Rs*

Ease of Administration

- Interviewer issues:
 - Does item require special training? Do *I*wers have trouble delivering the question, e.g., do they often stumble over certain words?
- Respondent issues
 - “Show cards” used to visually (persistently) display long lists of spoken (ephemeral) response options
 - If response involves difficult recall and event dating, *Rs* may be provided with calendars or other memory aids
- Dependencies among questions (less of a concern if questionnaire implemented on computer)
 - Skip patterns
 - Tailoring of questions, e.g., filling in *R*’s name from earlier question

Types of questions (from class 4)

1. Behaviors and facts vs. attitudes

“Have you ever been a ‘regular’ cigarette smoker, that is, at least one pack a week?”

“To reduce cigarette smoking some states have increased tax on the purchase of cigarettes. Do you favor or oppose this kind of tax?”

2. Sensitive (or threatening or stigmatized) vs. non-sensitive topics

“In the past 30 days, how often have you visited a website for sexually explicit material?”

“How many fluid ounces of plain drinking water did you drink yesterday from the tap or water fountain?”

3. Closed vs. open questions

“What political party do you identify with: Republican party, Democratic party, Independent or something else?”

“Why do you identify with that party?”

Open vs. Closed Questions:

- A continuum (from most open to most closed):
 - Questions that ask for multi-word response (*What is the most important problem facing the country today? Why do you oppose the death penalty?*)
 - Questions that imply a constrained set of answers (*Are you currently enrolled in a 401K plan with your employer?; How many brothers and sisters do you have? At what time do you usually leave for work?*)
 - Questions accompanied by a set of answer categories which the interviewer does not read to the *R* (“field coded items”)
 - Questions that include an explicit set of response categories which the interviewer is supposed to read to the respondent: Ordered categories/scales, unordered categories, check all that apply, etc.
- When survey researchers talk about open questions, they usually mean the first type
- Half-open questions: set of options plus “other (please specify)”

Open vs. Closed (continued)

- Open questions may elicit answers that would not have been produced if fixed options presented
 - Schuman & Presser (1981) asked about most pressing problem facing country in open or closed form
 - 22% of open responses concerned “energy shortage” which was major issue then
 - Closed form did not include “energy shortage” so could not be reported
- But specific options can remind people of possible answers they would not think of with open form
 - Shuman & Presser (1981) asked what attributes of a job people would prefer
 - Open version led predominantly to answers about “pay” though could not distinguish high from steady pay
 - Subsequently designed closed form that presented options for both: “work that pays well” vs. “work that is steady”

The Dimension of Judgment

- Most Qs (whether open or closed) ask *R* to place object of judgment (event, attitude) on a dimension
- Behaviors (events)
 - Occurrence Did event of this type happen?
 - Frequency How many times/how often did it happen?
 - Duration How long did the event last?
 - Elapsed time How long ago did it happen?
 - Quantities What is the total number of times?
- Attitudes
 - Favorability Bipolar
 - Strength/intensity Unipolar
 - Certainty Bipolar
 - Importance Unipolar

Questions about Facts and Behaviors

- Attributes
 - *“Is NAME male, female, transgender male, transgender female, non-binary, other, prefer not to answer?”*
 - *“Which category best describes your age?”*
- Behaviors
 - *“How often do you floss your teeth?”*
- Knowledge/awareness
 - *“Do you happen to know who wrote Huckleberry Finn?”*
 - *“Who is the chief justice of the US Supreme Court?”*

Recognition Versus Recall

- Recall: *“What print magazines have you read in the past week?”*
- Recognition: *“Which of the following print magazines have you read in the past week?”*
 - *Rolling Stone*
 - *New Yorker*
 - *National Geographic*
 - *People*
 - *Time*
- Latter approach is easier and reduces omissions but could also inhibit reports of items not on the list

Guidelines for Nonthreatening Questions about Behavior*

1. Find the simplest wording that will fulfill the question objectives
2. If the question is closed, include all reasonable options
 - *R* likely to infer that answer must be among the options provided
 - If no option corresponds to *R*'s circumstances, *Rs* unlikely to volunteer the actual answer to an *Iwer* or to write it on a paper questionnaire
3. Provide ample recall cues for retrospective Qs, especially cues that remind *R* of the context in which the relevant events were experienced (encoding specificity)

Were you the victim of assault, that is did someone use force against you...
with any weapons, for instance, gun, knife, scissors;
with anything like a baseball bat, frying pan, a chair, or stick;
by something thrown such as a rock, bottle, dish, hot liquid;
by grabbing, punching, choking, scratching or biting

*Adapted from Sudman & Bradburn (1982), *Asking Questions: A Practical Guide to Questionnaire Design*

Guidelines for Nonthreatening Questions about Behavior (continued)

3. Make the question as specific as possible; make it clear who and what is covered by the question and for what time period
 - *During the last week, when did you usually leave home to go to work?*
 - *vs. When do you usually leave home to go to work?*
 - More specific question invites answer based on recall – which may be desirable – rather than on impression, estimation, routines
 - If really want a very generic judgment, make this clear, e.g., “In general, how satisfied are you with your life?”
4. Pick a time period (“reference period”) appropriate for the topic
 - Hospital stays – six months
 - Doctor visits – a couple of weeks
 - Dietary recall – a couple of days?

Guidelines for Nonthreatening Questions about Behavior (continued)

5. Use words that virtually everyone understands; define special terms
 - *Do you procrastinate? vs. Do you put off until tomorrow things you could have done today – that is, procrastinate?*
 - Okay to use technical terms, jargon, or slang if *Rs* likely to know them
 - Even ordinary terms can be interpreted differently by different *Rs* (conceptual variability) so may need to be defined to standardize interpretation, “... and by ‘smoked a cigarette we mean that you took even one puff.”

Sensitive/Threatening Questions: Level of Sensitivity

- Socially undesirable (“contranormative”) behaviors
 - e.g., specific sex practices, illicit drug use, drinking and smoking, bankruptcy, criminal acts, criminal victimization, some medical procedures
 - *In the past 12 months, how many times have you used cocaine?*
- Socially desirable (“normative”) behaviors
 - e.g., voting, exercise, diet, getting children vaccinated, religious attendance
 - *On how many days did you exercise last week?*
- Sensitive questions vs. sensitive answers:
 - Some questions seen as invasion of privacy regardless of answer (e.g., income, religion, political affiliation; just raising the topic violates privacy)
 - Others are sensitive depending on the answer, e.g., “none – did not exercise on any days last week”

Measuring Threatening Behavior

- Approaches to writing Qs about sensitive topics (after Barton, 1958)
 - The casual approach: *Do you happen to have murdered your spouse?*
 - Softened wording: *As you know, many people have been killing their spouses these days. How about you? Do you happen to have killed your spouse?*
- Item count technique
- Evidence in favor of self-administration is very strong

Item Count Technique

- Two lists of behaviors created that are identical except one list contains target sensitive behavior and other does not
- “How many of the following have you done since January 1:
 - Bought a new car, traveled to England, donated blood, gotten a speeding ticket, and visited a shopping mall?”
 - Second list does not contain “gotten a speeding ticket”
- *Rs* presented with one of two lists
- Δ between counts from the two lists estimates prevalence of target behavior
- Evaluation to date suggests technique is effective and practical
- but does not provide data at individual respondent level

Guidelines for Threatening Questions about Behavior*

1. Use open rather than closed numerical questions for obtaining frequencies of sensitive behaviors
 - Allows *R* to answer without comparing themselves to others (e.g., may erroneously interpret middle range as the norm or average)
2. Use familiar words
 - “rape” vs. “sexually violate”
3. Use informants (proxy respondents)
 - Informants may be more willing to report sensitive behaviors (spouses may be better than parents)

*Adapted from Sudman & Bradburn (1982)

Guidelines for Threatening Questions about Behavior (continued)

4. Deliberately load the question

- *Everybody does it:* Even the calmest parents get angry at their children sometimes. Did your child(ren) do anything that made you yourself angry during the last seven days?
- *Presuppose the behavior:* When was the last time you smoked a cigarette?
- *Use authority to justify the behavior:* Many doctors now think that drinking can reduce heart attacks ...
- *Forgiving wording:* Did things come up which prevented you from voting or did you manage to vote this time?

Guidelines for Threatening Questions about Behavior (continued)

5. Starting with lifetime questions may help (*Have you ever, even once, used cocaine?*)
6. Embed the sensitive behaviors among other items
 - e.g., list abortion among other medical procedures
 - embedding in this way may reduce the threatening item's prominence
7. Self-administration!
 - Removing the interviewer from data collection eliminates or at least reduces their “social presence”
 - highly effective approach for increasing socially *undesirable*, and presumably more truthful, responses
 - Vast literature of mode comparisons and widespread use in practice supports this

Questions About Subjective States, i.e., Opinions, Attitudes

- Subjective states typically involve feelings about self or others, intuitions, and affect
 - Generally true value is unknowable and may vary depending on context
 - *Rs'* judgments range from highly subjective to relatively factual
 - Subjective: e.g., agree-disagree, favor-oppose,
 - Relatively factual, e.g., “Do you think the earth is flat?,” “Do you consider crime to be a big problem, some problem, or no problem at all?”)
- Most Qs about subjective states should address*:
 1. What is to be rated/evaluated (“attitude object”)?
 2. What dimension (response scale) is the rated object to be placed on?
 3. What characteristics of the dimension are offered to *R* (e.g., how many categories or scale points? Are they described verbally or numerically?)

* Fowler, 1995; chapter 3

What is to be rated (attitude object)?

- Attitude object should be defined
 - If different *Rs* rate (what are in effect) different objects than one another, results not comparable
 - e.g., “... and by *foreign policy* we mean”
- Assuring uniform interpretation across *Rs* is as important for questions about subjective (opinions) as objective (behaviors) phenomena
- Hubbard et al (2020) observed similar amounts of misunderstanding for *Qs* about opinions as behaviors

On what dimension should object be rated?

- Researchers ask *Rs* to place objects on dimensions such as
 - importance, satisfaction, size (extent), warmth, etc.
 - Often on symmetrical scale like agree-disagree or favor-oppose :
 - but this can promote *acquiescence* – tendency to select positive options irrespective of *R*'s actual subjective state
 - can create difficult response task, e.g., confusing to disagree with a negative statement
 - Also common, 101-point “feeling thermometer”
- Important to decide/determine if underlying dimension is symmetrical around zero or only positive?
 - Bipolar or Unipolar?

Bipolar Versus Unipolar Items

- Bipolar scales reflect two opposing alternatives; clear conceptual midpoint (even if not explicitly expressed)
 - *“Do you favor or oppose using public funds to support abortion services?”*
- Unipolar: varying levels of the same construct with no conceptual midpoint, zero at one end
 - *“I feel like I’m up against the world... Not at all, Sometimes, Often, All the time*

Balanced Versus Unbalanced Items

- Unbalanced: *“Are you in favor of the Affordable Care Act?”*
 - Yes = unambiguous
 - No = opposition, has no opinion, on the fence, has mixed feelings
- Explicit balance:
 - Favor or oppose, e.g., “Do you favor or oppose the Affordable Care Act?”
 - Allow or not allow
 - Permit or not permit
- Balanced arguments:
 - *“Some people think ... other people think ... what do you think...?”*

Number of Scale Points and Labels

- Number of scale points
 - General finding: seven-point scales lead to more reliable measures than fewer points and the benefits plateau at seven
 - Validity (e.g., criterion validity) also higher for moderate number (e.g., 7) than fewer or more scale points (but still contentious issue)
- Fully labeled scales: Labelling all points can increase reliability of responses but difficult to find more 5 or 6 labels, especially that *Rs* interpret consistently
- Some phenomena, e.g., warmth, lend themselves to continuous measurement which generally entails many points
 - Feeling thermometers, usually 101 points (0 – 100)
 - Visual analog scales, often implemented with sliders in web surveys

Partially vs. Fully Labeled

End-Point Only

Extremely Satisfied



Extremely Dissatisfied



Fully Labeled

Extremely Satisfied

Somewhat Satisfied

Neither Satisfied nor Dissatisfied

Somewhat Dissatisfied

Extremely Dissatisfied

Fully Versus Partially Labeled Scales

- Alwin and Krosnick (1991): calculate average reliability for two types of labeling
- Significant improvement in reliability with verbal labeling

Labeling	Number of Items	Adjusted Mean Reliability
Full verbal labels	5	.783
Endpoints only	8	.570

- See also Krosnick & Fabrigar, 1997; Alwin, 2007; Saris & Gallhofer, 2007

Rating Versus Ranking

- Rating (evaluating each item on same scale)

“How likely, on a scale of 1-5, would you be to buy each type of chocolate

vs. Ranking (ordering all options relative to one another on scale)

“Please assign a value of 1 to the type of chocolate you are most likely to buy, a 2 to the one you are second most likely to buy, etc.

- Ranking is a more difficult cognitive task:
 - must keep in mind which objects have been ranked and what their rank is
- Average ratings provide aggregate rankings
- Ranking in online (as opposed to paper) questionnaires can enable R to drag and drop objects, eliminating working memory demands

Guidelines for Attitude Questions*

1. Make sure that the attitude objects are clearly specified

- *“Do you think the government is spending too little, about the right amount, or too much on public assistance?”*
- What is public assistance? What level of government? How much is it currently spending?

2. Avoid double-barreled items (that tap two attitudes at once)

- e.g., *“The instructor was fair and rigorous in grading assignments.”*
- What if the instructor was rigorous but not fair?

*Adapted from Sudman & Bradburn (1982), *Asking Questions: A Practical Guide to Questionnaire Design*

Guidelines for Attitude Questions (continued)

3. Measure the strength of the attitude

- Incorporate the strength dimension into the attitude question itself (*How do you feel about the death penalty for convicted murderers. Do you strongly favor it, favor it, neither favor or oppose it, oppose it or strongly oppose it?*)
- Add a separate question to assess strength (“*How do you feel about the death penalty for convicted murderers. Do you favor it, oppose it, or do you neither favor nor oppose the death penalty?*” For those in favor: “*Do you favor it strongly or mildly?*”)
- if it is likely that in practice, you will collapse “strongly” and “somewhat” responses, may reduce burden on *Rs* if you only ask provide one option on either side of midpoint: “favor,” “neither favor nor oppose,” “oppose”

Guidelines for Attitude Questions (continued)

4. Middle options, Don't Know*:

- If these are included, they will be selected more than they would be volunteered (in an interview or on a paper questionnaire)
- *Rs* may select DK or middle options to reduce effort – to satisfice – or because don't feel they know enough for opinion to warrant reporting
- Evidence about benefits of including DK is mixed: some studies press *Rs* who initially select DK option to select substantive option, increasing predictive validity (Visser et al. 2000) but other studies report no impact of including DK (Krosnick et al., 2000)
- Decide on case-by-case basis after careful pretesting

*Not the same: possible for *R* to know that they neither favor nor oppose

Limitations of Guidelines

- For every rule, there are exceptions
- Sometimes the guidelines conflict
- Design most likely to be effective when it involves
 1. principles and theory about response process
 2. pretesting

Pretesting

- No matter how well designed a question may be, it's hard to know on purely intuitive grounds how respondents will react to it
- Best practice is to test questions empirically before administering them in production surveys
- Several widely used pretesting methods: Expert Review, Behavior Coding, Interviewer Debriefing, Cognitive Interviewing
- Cognitive interviewing particularly effective at exposing response difficulty, especially involving comprehension
- Behavior Coding especially effective at exposing problems in I-R interaction and problems administering questions
- Iterative design: design => test => redesign => retest (repeat as necessary)

Cognitive Interviewing

- Much better to conduct even a few cognitive interviews than not
 - Can uncover problems before data collection begins, allowing designers to revise questions
 - Can immerse researchers in questionnaire design more deeply
 - Can provide evidence to persuade others that revisions are needed
- But
 - *Rs* may indicate they have experienced a problem when they have not (Conrad & Blair, 2009) – training *Interviewers* to avoid specific probes can help
 - More cognitive interviews likely to uncover more problems than fewer interviews (Blair & Conrad, 2011) – more interviews requires more resources